Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: IV Semester B.E. EEE
Course Code & Title: IC8451 Control Systems
Name of the Faculty member: Mr. A. Guna, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit V / Concept of Controllability and Observability
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO 24
- **Activity Chosen: Flipped Classroom**
- **Justification:**

Flipped classroom is a "pedagogical approach in which direct instruction moves from the group learning space to the individual learning space, and the resulting group space is transformed into a dynamic, interactive learning environment where the educator guides students as they apply concepts and engage creatively in the subject matter"

Time Allotted for the Activity: 40 Minutes

- **Details of the Implementation:**

Number of Students – 2

Name of the students – S.Vijay Anandh, J. Senthil Murugan

Photographer: one student –Mr.R.Sangiliguru(interested in photography)

Reporter: Myself

✓ Four Students were asked to form two teams to present the concept ofcontrollability and observability to their classmates

✓ I shared the material related to the topic to the students and they presented to their classmates

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO4	PO10	PO12	PSO3
C214.5	3	3	3	1	1	2

- **PO / PSO mapped:**

PO	Justification for correlation
PO1	Students apply their fundamental engineering concept to understand the concept of observability and controllability
PO2	Students apply their mathematical concepts such as matrices to find the controllability and observability of the given system
PO4	Students analyze the controllability and observability of the given system
PO10	Students listen to the other student presentation and his way of explanation
PO12	Students apply the controllability and observability concepts for all the mechanical system
PSO3	Students will be able to develop the required hardware skill for industrial automation system

- **Images / Screenshot of the practice:**

❖ *Benefit of the practice:*

- This practice improves the peer coaching concept among the students
- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

References:

1. https://www.panopto.com/blog/7-unique-flipped-classroom-models-right/

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: IV Semester B.E. EEE
Course Code & Title: IC8451 Control Systems
Name of the Faculty member: Mr. A. Guna, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit I / Basic Elements in Control System – Open and closed loop Systems
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO1
- **Activity Chosen: One-minute paper**
- **Justification:**
  - The basic block diagram of control system has various components and the students understand the type of system through various examples. After teaching the concept, I thought of conducting this activity for making the students to give the difference between open and closed loop system which enhance the learning level of the students and as a teacher I can evaluate the understanding level of the students**.**

Time Allotted for the Activity: 5 Minutes

- **Details of the Implementation:**

After teaching the concept, students were asked to take a sheet of paper and write the concepts they understood about open and closed loop system.

Total Strength is **30**

Reporter: **Myself**

At the end the Class (Last 5 minutes)

- ✓ I asked the students to think about difference between open and closed loop system
- ✓ Then I told them to write as much as they can within a short period of 1 time.
- ✓ Finally, I collected the papers from the students

- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO8	PO10	PO12	PSO3
C214.1	**3**	**3**	**1**	**1**	**1**	**2**

- **PO / PSO mapped:**

PO	Justification
PO1	Students will have fundamental Engineering Knowledge to understand the examples of open and closed loop system
PO2	Students will identify the mathematical, engineering and other relevant knowledge that applies to derive the block diagram of the system.
PO8	Students will submit their One-minute paper without copying anyone
PO12	Students will be able to submit a clear, well written statement.
PSO3	Students will be able to represent a mechanical system in a block diagram

- **Images / Screenshot of the practice:**

❖ *Reflective Critique:*

1. **Pre-implementation Reflection:**

   Benefits:

   - Students can able to attend the question even if the questions are in indirect form.
   - Students can able to write the concepts in examination without any confusion.

   Challenges: Time utilization for conducting activity.

2. **Post-implementation Reflection:**

   Benefits: Students understood the concept which was reflected from their answers during the next class.

   Challenges: Late bloomers had some difficulties in answering the questions during the short period of time.

❖ *Benefit of the practice***:**

- Most of the students able to recollect the concept quickly whenever asked
- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 80% of students answered.

References:

❖ https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-one-minute-paper

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE
Course Code & Title: IC8451 Control Systems
Name of the Faculty member (s): Mr. A. Guna, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit II /
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO8
- **Activity Chosen:** MATLAB Simulation
- **Justification:**
  MATLAB provides you a huge number of toolboxes to plot graphs, visualize complex datasets. Students use MATLAB to perform complex scientific calculations, it also lets the students to visualize and implement complex mathematical functions. With the help of MATLAB simulation Students can determine the time domain specification of the given system in an easy way.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
  - ✓ All the students were taken to Control and Instrumentation Lab and the procedure to simulate the given system is explained to them
  - ✓ Students individually simulated the time response of the given system
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO4	PO5	PSO1
CO2	3	3	2	2	1

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO4	PO5	PSO1
	3	3	2	2	1
Justification for correlation	Student applying their engineering knowledge to understand the concept of time domain analysis of a given system	Students apply mathematical concepts to determine the time domain specification of the given system	Students analyze the response of the system with the desired response and draws necessary conclusion	Students use MATLAB simulation software for determining time response of the system and their proficiency in MATLAB simulation is improved	Student analyze the performance of the given system using MATLAB software Package

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***
    Students find this simulation is the easier way to determine the time domain specification of the given system.
    Students also assured to continue the simulation using Online MALAB Simulation

  - ***Benefit of the practice:***
    The time response of a higher order system or any complicated system can be determined using MATLAB simulation
    Student understood the time domain specifications clearly and it is reflected during the discussion session.

  - ***Challenges faced in implementation:***
    Few late bloomer students felt difficult in coding and simulating the transfer function

References:

https://in.mathworks.com/support/learn-with-matlab-tutorials.html

Signature of Faculty Member **HOD**

Department of Electrical and Electronics Engineering
Academic Year 2021 – 2022 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. EEE
Course Code & Title: IC8451 Control Systems
Name of the Faculty member (s): Mr. A. Guna, AP/EEE

Innovative Practice Description

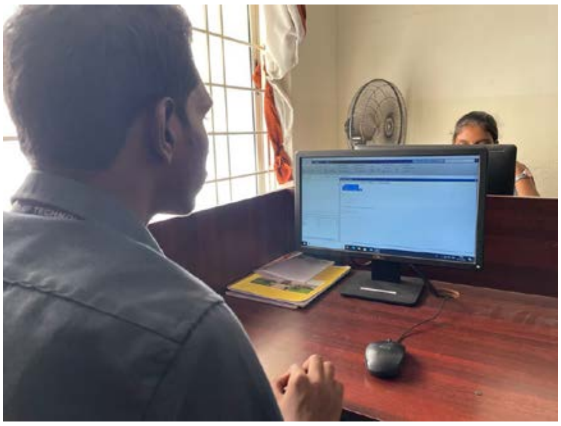
- **Unit / Topic:** Unit III /Bode Plot
- **Course Outcome:** CO 3
- **Topic Learning Outcome:** TLO13
- **Activity Chosen:** MATLAB Simulation
- **Justification:**
  MATLAB provides you a huge number of toolboxes to plot graphs, visualize complex datasets. Students use MATLAB to perform complex scientific calculations, it also lets the students to visualize and implement complex mathematical functions. With the help of MATLAB simulation Students can determine the stability of the given system using Bode Plot in an easy way.
- **Time Allotted for the Activity:** 40 minutes
- **Details of the Implementation:**
  - ✓ All the students were taken to Control and Instrumentation Lab and the procedure to simulate the given system is explained to them
  - ✓ Students individually simulated the Bode plot of the given system
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO4	PO5	PSO1
CO2	**3**	**3**	**2**	**2**	**1**

- **PO / PSO mapped:**

Innovative practice	**PO1**	**PO2**	**PO4**	**PO5**	**PSO1**
	3	**3**	**2**	**2**	**1**
Justification for correlation	Student applying their engineering knowledge to understand the concept of stability analysis of a given system	Students apply mathematical concepts to determine the stability of the given system	Students analyze the stability of the system and draws necessary conclusion	Students use MATLAB simulation software for determining time response of the system and their proficiency in MATLAB simulation is improved	Student analyze the performance of the given system using MATLAB software Package

- **Images / Screenshot of the practice:**



- **Reflective Critique:**
  - ***Feedback of practice from students and other stakeholders:***

    Students find this simulation is the easier way to determine the stability of the given system.

    Students also assured to continue the simulation using Online MALAB Simulation
  - ***Benefit of the practice:***

    The stability of a higher order system or any complicated system can be determined using MATLAB simulation

    Student understood the Bode plot concept clearly and it is reflected during the discussion session.
  - ***Challenges faced in implementation:***

    Few late bloomer students felt difficult in coding and simulation

References:

https://in.mathworks.com/support/learn-with-matlab-tutorials.html

Signature of Faculty Member **HOD**